COLIMITS OF QUASI-COHERENT SHEAVES ON AN AFFINE SCHEME ARE QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1658-1672, following the statement at 1651-1656.

Editorial note. The direct-sum case is displayed in full; the colimit sentence 'A similar argument works for general colimits' is the same displayed computation with localization cocontinuous (left adjoint -f), a standard ingredient; flagged here rather than treated as an omission of new content.

Allowed standard prerequisites (editorial).

- Equivalence of quasi-coherent sheaves with modules on affines. lemma-equivalence-quasi-coherent (same chapter, cited at ~1660). Writes each F_i as tilde(M_i).

Source reference. Original lemma 01IB: sources/stacks-schemes/pinned-original-source.tex, source lines 1607–1607. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X = \mathrm{Spec}(R)$ be an affine scheme. The direct sum of an arbitrary collection of quasi-coherent sheaves on X is quasi-coherent. The same holds for colimits.*

Proof. Suppose $\mathcal{F}_i$, $i \in I$ is a collection of quasi-coherent sheaves on X. By Lemma 01IB above we can write $\mathcal{F}_i = \widetilde{M_i}$ for some R-module M_i. Set $M = \bigoplus M_i$. Consider the sheaf $\widetilde{M}$. For each standard open $D(f)$ we have

$$\widetilde{M}(D(f)) = M_f = \left(\bigoplus M_i\right)_f = \bigoplus M_{i,f}.$$

Hence we see that the quasi-coherent $\mathcal{O}_X$-module $\widetilde{M}$ is the direct sum of the sheaves $\mathcal{F}_i$. A similar argument works for general colimits. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.